

Adults' use of latent causes for reasoning about Common Causes and direct causes for reasoning about Causal Chains

Background

This study builds on a prior pre-registered experiment conducted in our lab (Study 1, June 2025: [link redacted for future double-blind review]) which investigated the hypothesis that adults use two representational spaces to organize mental, physical and bodily events. One space groups events based on underlying latent factors, and another space describes how specific events are directly causally connected to each other. Specifically, study 1 found that: (1) adults represent 3 latent factors namely “the mind”, “the mechanical body” and “the biological body” that organize mental events, actions, and physiological events respectively, and (2) adults hold a distinct representation of how events within and across the latent factors causally relate to each other.

An open question regarding the two representational spaces is what kind of content they articulate. One possibility is that these two spaces are intuitive causal-explanatory framework theories about how other people work. Prior work has shown that intuitive theories are common-sense, abstract, causal-explanatory frameworks that help us navigate the world (Gerstenberg & Tenenbaum, 2017; Gopnik & Wellman, 1994). They help people reason about events from different domains: Intuitive psychology, for example, conceives of actions as causally connected to mental states, which allows people to explain, predict and intervene on other agents' minds and behaviors. Intuitive theories also help people reason about events within the same domain: children's intuitive understanding of the biological domain helps them explain, predict and plan interventions on bodily states to prevent states of illness and disease, and to infer that members of living species that share internal structure—not superficial perceptual features—will have similar properties (Degn et al., 2025; Gelman & Markman, 1987).

Here, we propose to test the hypothesis that the judgments that we measured in Study 1 constitute two distinct intuitive theories that help us make sense of other agents as mental beings, physical actors, and living systems in terms of (1) Latent Causes and (2) Direct Causes. If the representational spaces measured in Study 1 constitute intuitive theories, then even when not asked to, adults should recruit these beliefs for prediction, explanation, counterfactuals, and intervention. If these two spaces represent two distinct ways of reasoning causally about other agents, then we should be able to predict when people will use beliefs about Latent Causes vs Direct Causes.

In summary, in the current work we propose to test for the functional roles of the Latent Causes and Direct Causes frameworks as distinct intuitive causal theories in common-sense social cognition.

General Alternative Hypothesis

The two frameworks (Latent Causes and Direct Causes) support common-sense reasoning: We hypothesize that the two causal frameworks measured in Study 1 are the basis for our intuitive reasoning within and across the domains of mind, action and body. We predict that adults will spontaneously make use of these representations for inference, explanation, intervention, and counterfactual reasoning.

The two frameworks do not support common-sense reasoning: An alternative hypothesis is that while adults can report their beliefs about how mental events, actions, and physiological events are organized when explicitly asked to (like in Study 1), they do not make use of these beliefs during everyday social cognition.

Current Study

Prior work has demonstrated that adults and children can represent and reason about a variety of causal structures including causal chains, common-cause and common-effect structures ([Gopnik et al. 2004](#); [Schulz et al. 2007](#); [Pearl 2000](#); [Rottman et al. 2012](#)). In the current work we aim to explore whether adults use representations of direct causes and latent causes to reason about causal chain and common cause structures, respectively.

In the prior study from our lab (Study 1), adults were asked whether one event can make a second event happen, and reported their responses on a sliding scale. One open question is whether adults who deny a direct causal link between two events (A does not cause C) can still expect them to be connected indirectly, via third event that is both an effect of the first event and a cause of the second event, thus forming a causal chain (A can cause B, and B can cause C).

Additionally, prior work also shows that children and adults view natural kind categories, including the abilities of other agents, as consisting of clusters of correlated properties arising from an underlying latent cause (an unobservable essence). These categories support inductive inferences about novel instances and shape our folk taxonomy ([Gelman and Markman 1986](#); [Gelman and Wellman 1991](#); [Neufeld 2022](#); [Weisman et al. 2021](#); [Kim et al. 2019](#); [Hatano and Inagaki 1994](#)). In our lab's prior study (Study 1) adults sorted phrases into events of the mind, actions, and events of the body; do these three clusters reflect three latent causes? Here we use a paradigm inspired by the literature on essentialist reasoning about natural kinds to test whether adults expect events that group together to have a common latent cause (a biological substrate).

We hypothesize that direct causes are relevant for completing a causal chain between two causally distant events (i.e. for figuring out which B event completes the chain $A \rightarrow B \rightarrow C$), and latent causes are relevant for inferring a missing effect in a common-cause structure (i.e. if two effects, A and B, are caused by the same variable, what other effects we should expect). Thus, we designed a causal chain task and a common cause task: in the causal chain task, participants judge which of two candidate middle events best completes a causal chain between two causally distant **target** events (i.e. causally distant per Study 1

responses); either a **causal item** (i.e. an ability which is **caused by** the first and **causes** the third target ability) or a **similar item** (i.e. an ability which is “similar” to the first and third target abilities owing to having a common cause with both targets).

We constructed a stimulus set based on the items and results of Study 1. The set consists of **9** triads of items, each triad consisting of a **first target** item (such as “sit down”) and a **third target** item (such as “jump up and down”) and a choice set of two items for the **middle item**. The two items are (1) a **similar** item: an item that was most **similar** to both the first and third target items (i.e. close by in the Sorting Task RDM to both the first and third items on average) (such as “take a walk”), and (2) a **causal** item: an event that the first target was a plausible cause for, and was a plausible cause for the third target, while also being distant in similarity space (i.e. a close-by causal successor in the Causal Task RDM but far away in the Sorting Task RDM) (such as “experience pain”). We constrained the similar option to be from the same domain as the target (this was already true for 8/9 target items), and the causal option to be from a different domain (this was already true for 9/9 target items). The full set of 9 item triads are listed in Table 1.

Table 1: Items for the current study based on group-averaged distances from two separate samples (total N = 151). The first three columns list the target events and their domains. The remaining two columns list the two options (the “causal choice” vs “similar choice”) associated with that target, and their similarity and causal distance from the target.

	Target items presented in a causal chain vs common cause context		Middle item in forced choice response	
domain	First target	Third target	Causal choice	Similar choice
mind	see something	hear something	take a walk	remember something
	choose what to do	hear something	take a walk	see something
	remember something	hear something	feel scared	think about something
action	sit down	jump up and down	experience pain	take a walk
	jump up and down	sit down	get tired	take a walk
	take a walk	reach for something	see something	jump up and down
	get tired	feel scared	choose what to do	become hungry
body	become hungry	feel scared	think about something	get tired
	get sick	become hungry	remember something	experience pain

Hypothesis

We hypothesize that people can use representations of direct causes (i.e. whether A can make B happen) to construct a coherent causal sequence of events, and representations of latent causes (i.e. whether A and B are both caused by a common variable) to infer which abilities a newly discovered gene controls. Specifically, when people are asked to infer which

event best completes a three-event sequence (e.g. first event = sit down; third event = jump up and down; causal choice = experience pain; similar choice = take a walk), they will select the direct cause (e.g. someone sat down, which made them experience pain, which made them jump up and down) and they will select the item which shares an underlying cause with the first and third event when predicting which other ability is controlled by the same genetic substance (e.g. if a gene controls whether a novel creature can sit down and jump up and down, it also likely controls whether they can take a walk).

We predict that people will be (1) more likely to select the causal item in the Causal Chain than the Common Cause condition, they will be (2) more likely than chance to select the causal item in the Causal Chain condition, and they will be (3) more likely than chance to select the similar item in the Common Cause condition.

Dependent variable

On each trial, the choice between the **causal** and **similar** item.

Independent variable

Whether the task involves selecting the missing event in a **Causal Chain** or the missing effect in a **Common Cause** structure.

Procedure

Subjects will be randomly assigned to two conditions. Subjects in each condition will be presented with the following instruction page:

Causal Chain condition: Welcome! In this game you will be asked to consider a sequence of three events that occurred. The events are connected, such that the first event made the second event happen, which made the third event happen. For example you might be shown: someone did X, which made them do Y, which made them do Z. On each trial you will see events in the first and third positions. Your job is to indicate the most likely middle event that will complete the chain.

Common Cause condition: Welcome! In this game you will be asked to imagine that you are a scientist studying the biological basis of behaviors in new creatures. Some of the behaviors are related, in that they all arise from the same genetic material. For example you might be shown: gene A gives creatures the capacities X, Y and Z. On each trial you will see the first and third capacity. Your job is to indicate the most likely middle capacity that would also arise from the gene.

The next pages will present 15 trials in each condition, randomly interspersed. Across both conditions, trials will consist of a test question and two alternatives in a forced-choice paradigm. The trials will read as follows:

Causal Chain condition: Which phrase best fits the blank?

Someone {**first event** e.g. sat down}, which made them _____, which made them {**third event**, e.g. jump up and down}.

Choices: {**causal choice** e.g. experience pain}; {**similar choice** e.g. take a walk}

Common Cause condition: Which phrase best fits the blank?

You discovered a gene, UT9025*, which controls whether a species of creature can {**first event** e.g. sit down}, _____, and {**third event** e.g. jump up and down}.

*UT9025 is a randomly selected alphanumeric sequence from the randomly generated set: HFE1A, BRCA1, UT9025, CD8R1, ZBT7A, VF5013, APP125, FTR540, TRE891. These will be randomly assigned to each trial without replacement.

Each of the 15 targets will be shown once per condition. Below the test question will be an option between a similar choice and a causal choice. The left-right positions of the choices will be randomised across trials.

There will be two randomly interspersed attention checks, and participants will be excluded if they failed at least one attention check. The attention checks will be as follows:

Causal Chain condition: For this question, please select 'jump up and down'.

Choices: Jump up and down, Choose what to do.

Common Cause condition: For this question, please select 'choose what to do'.

Choices: Jump up and down, Choose what to do.

After the 30 trials, subjects will be presented with a debrief followed by a demographics questionnaire.

Analysis Plan

To test for our 3 predictions—namely, that people will be (1) more likely to select the causal item in the Causal Chain condition than the Common Cause condition (2) more likely than chance to select the causal item in the Causal Chain condition, and (3) more likely than chance to select the similar item in the Common Cause condition—we will use a mixed effects logistic regression using the lme4 package (Bates et al., 2015) in R. The model specification will be: **response ~ condition + (1 | subject_id), family = binomial(link = 'logit')**. We will set the reference group to be the Causal Chain condition, thus the “intercept” will represent the log-odds of selecting the causally relevant item in the Common Cause condition, the slope coefficient will capture the log-odds difference between the Causal Chain and Common Cause conditions, and the sum of the intercept and the condition coefficient will give the log-odds of selecting the causally relevant item in the Causal Chain condition. We will confirm our first prediction if the slope coefficient is significant, and we will confirm our second and third predictions by running 'allEffects()' from the *effects* package, which will generate confidence intervals that show whether each condition mean is different from chance.

Our threshold for statistical significance will be $p = .05$, two-tailed, and we will use the `check_model()` function from the `performance` package (Lüdtke et al., 2021) to conduct quality assurance.

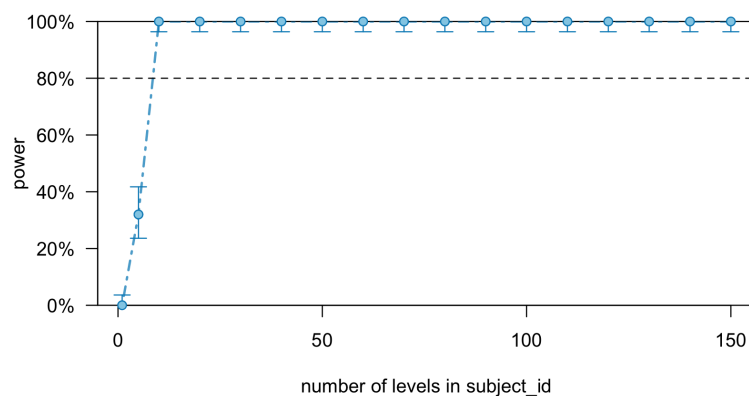
Sample Size and Stopping Rule

a) Power analysis for Confirmatory Analysis

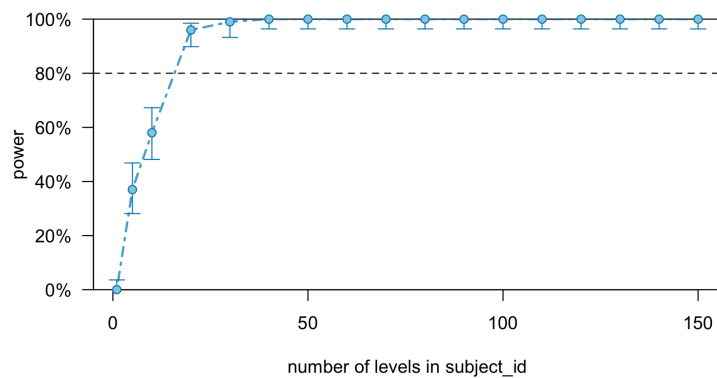
We used the R package *simr* to estimate the power curves for replicating the 3 predicted results. For power analysis purposes, we ran three models targeting each prediction separately: a combined condition model for detecting the condition difference across the Causal Chain and Common Cause condition, a Causal Chain-only model for detecting if the mean proportion responding with the causal item is above chance, and a separate Common Cause-only model for detecting if the mean proportion responding with the similar item is above chance.

Based on a pilot study ($N = 105$), the sample size needed to achieve 80% power to replicate the condition effect is $5 < N < 10$, the sample size required to replicate the Causal Chain effect is $10 < N < 20$ and the sample size to replicate the Common Cause effect is $5 < N < 10$ (Figure U3).

(i) Condition difference effect



(ii) Causal Chain condition effect



(iii) Common Cause condition effect

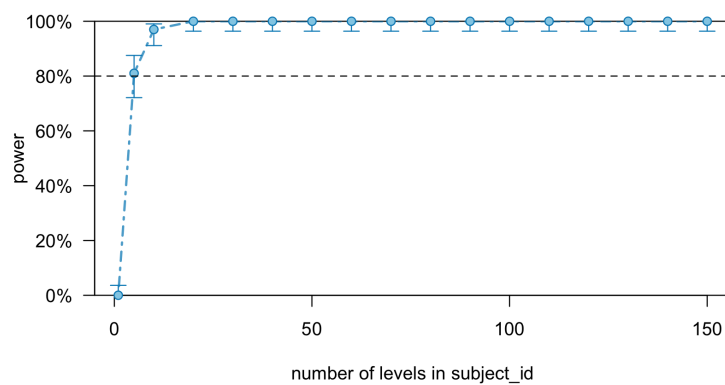


Figure U3(a-c): Power curve showing the power that would be achieved (y-axis) using different sample sizes (x-axis) to replicate the fixed effects for the responses $\sim$ condition + (1 | subject_id) models with different datasets. (a) Shows power curve for the model of the condition difference between Causal Chain and Common Cause conditions (b) shows power curve for the Causal Chain-only model, and (b) shows power curve for the Common Cause-only model.

b) Stopping rule and exclusions

Considering the largest sample required to replicate all effects ($N \approx 20$), and that the effect size in the pilot could be inflated due to a small sample size, we conservatively estimate a target sample of 100 participants. To account for a <1% exclusion rate from the pilot study, we will conservatively collect data from 105 participants, prior to exclusions.

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